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Refolding, purification, and activation of miniplasminogen and microplasminogen isolated from *E. coli* inclusion bodies

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Abstract

Two des-kringle derivatives of human plasminogen, microplasminogen and miniplasminogen, have been expressed at high levels as inclusion bodies in *Escherichia coli* using a T7 expression system. In each case, the isolated inclusion bodies were refolded and purified. A final yield of ~10% of total refolded protein was observed in each case. Both refolded molecules were successfully activated to their functional forms, microplasmin and miniplasmin, by the plasminogen activator urokinase. The kinetic properties of the refolded microplasmin and miniplasmin were comparable to full length, native plasmin.
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Keywords: Miniplasmin; Microplasmin; *E. coli* expression; Inclusion bodies; Refolding; Activation

Plasmin (Plm)[2] is a serine protease with several important physiological roles, including dissolving blood clots. Plasmin is normally found circulating inertly in the blood in its zymogen form, plasminogen (Plg): a 791 amino acid single-chain glycoprotein (Mr ~ 87 kDa) that possesses the capacity to bind to newly formed blood clots. Its activation is effected by digestion of the peptide bond between arginine 561 and valine 562 ($R^{561}\blacktriangledown V^{562}$) by tissue plasminogen activator (tPA) or urokinase (uPA) trapped in the blood clot. Cleavage at this peptide bond transforms the single chain protein into two separate subunits interconnected by two essential disulfide bridges. The A chain of the Plm molecule consists of five triple-loop disulfide kringle (Kr) domains (approximately 78–80 amino acids each), while the B chain contains a "linker" region of 20 amino acids and a serine protease domain (approximately 228 amino acids). The newly formed plasmin actively digests the fibrin in the clot, thereby dissolving it.

Through laboratory manipulations, two des-kringle variants of plasminogen with potential pharmacological application, miniplasmin (miniPlm) and microplasmin (μPlm), were elucidated two decades ago. MiniPlm consists of only the kringle-5 domain, the linker, and the serine protease domain, while μPlm consists of only the linker and serine protease domain. MiniPlm is produced by digestion of full length plasmin with neutrophil elastase [1,2], which cleaves specifically at the peptide bond between valine 441 and valine 442 ($V^{441}\blacktriangledown V^{442}$). μPlm was initially produced by pH 11 base-mediated cleavage of full length plasmin at the peptide bond between arginine 530 and lysine 531 ($R^{530}\blacktriangledown K^{531}$) [3,4]. In the laboratory, μPlm and miniPlm can be generated from their zymogen precursors, microplasminogen (μPlg) and miniplasminogen (miniPlg), respectively, by cleavage at the same peptide bond as was described for plasminogen. Similarly, two separate subunits interconnected by the same two disulfide bridges are generated in each case. Removal of either one or both of these two

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[2] *Abbreviations used:* Plg: plasminogen; Plm, plasmin; μPlg, microplasminogen; μPlm, microplasmin; miniPlg, miniplasminogen; miniPlm, miniplasmin; Kr, kringle domain; tPA, tissue plasminogen activator; uPA, urokinase GSH, reduced glutathione; GSSG, oxidized glutathione; DTT, dithiothreitol; β-ME, β-mercaptoethanol; SEC, size exclusion chromatography; SP IEX, sulfopropyl ion exchange chromatography; IPTG, isopropyl-thio-galactoside; MW std, molecular weight standard; SDS–PAGE, SDS–polyacrylamide gel electrophoresis.

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disulfide bridges by mutagenesis renders the activated serine protease domain non-functional [5].

Recent advances in thrombolytic therapy have, in part, been driven by significant technical advances in catheter design and delivery, permitting local drug delivery directly into a clot. These innovations are particularly relevant to treatment of peripheral arterial occlusion disease (PAOD). In landmark studies, Novokhatny, Marder, and coworkers [6,7] were able to show, in an animal model of abdominal aorta thrombosis and in *in vitro* models, that serum-derived plasmin was equivalent to recombinant tissue plasminogen activator (tPA) in dissolution of the clot during unimpeded blood flow. More importantly, plasmin was clearly superior under conditions of restricted blood flow where Plg substrate was not replenished. Furthermore, plasmin caused dramatically less bleeding at a hemostatically stable ear puncture site distant from the thrombus site. The advantage of plasmin was realized because of its remarkably short half-life relative to tPA (0.02 s vs. ~15 min *in vivo*) [8].

These studies "opened the door" to similar studies demonstrating the utility of μPlm for treatment of PAOD and ischemic stroke. Nagai and coworkers [8] compared local catheter mediated delivery of serum-derived plasmin, recombinant μPlm isolated from yeast *Pichia pastoris*, and recombinant tPA and showed similar efficacy in clot dissolution using a rabbit extracorporeal loop thrombolysis model to simulate PAOD and a mouse middle cerebral artery model to simulate ischemic stroke.

Due to its fragile, complex structure, plasmin has never been produced at a pharmaceutical scale and is typically isolated from blood, and has never been recombinantly produced at high yield, nor refolded from inclusion bodies. Its lability at every stage of handling—from isolation to purification, formulation, and administration—has impeded its commercial development. While Nagai and coworkers have produced μPlg recombinantly at high yield using *P. pastoris*, there have been no reports of high yield recombinant production of miniplasmin. We report here the successful refolding of both μPlg and miniPlg from *Escherichia coli* inclusion bodies, with subsequent activation to functionally active μPlm and miniPlm.

Materials and methods

Construction of μPlg and miniPlg expression vector

Synthetic cDNAs encoding μPlg and miniPlg optimized for expression in *E. coli* were synthesized by CODA Genomics (Laguna Hills, CA). The protein sequences used to generate the cDNAs were obtained from the protein sequence for full length human plasminogen (Accession No. P00747) from the ExPASY server (www.expasy.org) of the Swiss Institute of Bioinformatics. The optimized cDNA encoding μPlg was inserted into the unique *Bam*H1/*Xho*1 sites of a custom pET11a expression vector with a modified expanded multiple cloning site, creating the plasmid pET11a-μPlg. The optimized cDNA encoding miniPlg was inserted into the same vector but at the unique *Nde*I/*Bam*H1 sites to create expression vector pET11a-miniPlg. The final clones were DNA sequence verified (MWG, High Point, NC).

Expression of μPlg and miniPlg in E. coli

Expression of μPlg and miniPlg was carried out using the auto-induction method developed by Studier at Brookhaven National Laboratory [9]. *E. coli* BL21 (DE3) cells were transformed with either pET11a-μPlg or pET11a-miniPlg and plated onto PA-0.5G/Ampicillin plates and incubated for 16 h at 37 °C. One colony was used to inoculate 50 ml PA-0.5G liquid media and incubated in a shaking flask at 37 °C for 16 h to an OD_{600} of 5–6 and immediately stored at 4 °C to create a working stock. The working stock (0.5 ml) was used to inoculate 500 ml ZYP-5052 liquid media and incubated 16 h at 37 °C, 300 rpm to an OD_{600} of 8–10 in a 2.8 L Fernbach flask. For larger scale production, 6–12 L was produced in multiple 2.8 L Fernbach shaker flasks.

For the data in Fig. 1 only, 3 ml cultures of pET11a-μPlg/BL21(DE3) and pET11a-miniPlg/BL21(DE3) were grown in LB amp broth to OD600 = 0.6. IPTG was then added at a final concentration of 1 mM and the culture was induced for 3 h at 37 °C. The samples were then spun down and the pellets were suspended in SDS running buffer with added 100 mM β-ME and loaded onto SDS–PAGE for analysis. Controls without IPTG were run in parallel.

Isolation and solubilization of inclusion bodies

After induction, the cells were harvested and centrifuged at 7000 rpm, 4 °C for 10 min. The pellet was then resuspended in 50 ml of 50 mM Tris, 25% sucrose, 1 mM EDTA, and 10 mM DTT, pH 8.0. Lysozyme (100 mg) was added and the solution was stirred for 30 min at room temperature. A buffer (125 ml) containing 20 mM Tris, pH 7.5, 1% NaDeoxycholate, 1% Triton X-100, 10 mM NaCl, and 10 mM DTT was added and the solution was stirred for an additional 30 min and then frozen at −85 °C for 20 h. The solution was then thawed in a 37 °C water bath for 3 h and homogenized using a Branson ultrasonic horn at 70% amplitude, 30 s pulse, 10 s pause for 6 cycles. The solution was brought to 1 L with 50 mM Tris, 100 mM NaCl, 0.5% Triton X-100, 1 mM EDTA and 1 mM DTT, pH 8.0 (Triton Solution), and stirred at 800 rpm, 4 °C for 1 h. The inclusion bodies were then centrifuged at 7000 rpm, 4 °C for 1 h, and the resulting pellet was resuspended with 1 L of Triton Solution and stirred again at 800 rpm, 4 °C for 1 h. The cycle of centrifugation and resuspension in Triton Solution was repeated 3 more times followed by four washes with 50 mM Tris, 1 mM EDTA, and 1 mM DTT (Tris Solution). After the last wash with Tris solution, the pellet was solubilized in 10 ml 8 M urea, 100 mM Tris, 1 mM glycine and 100 mM β-ME, pH 10.5, and stirred for 16 h at 300 rpm, 4 °C. The solubilized inclusion bodies were then ultracentri-

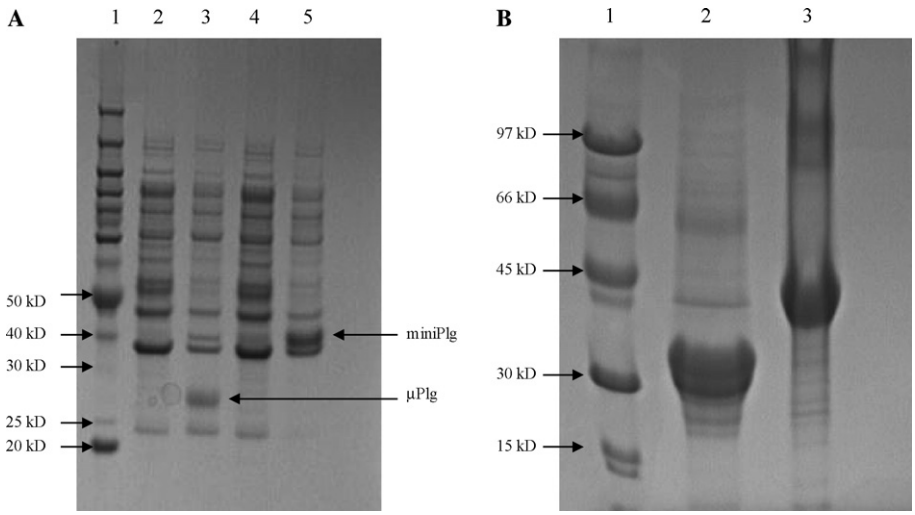


Fig. 1. (A) SDS–PAGE analysis (with β-ME reduction) of whole cell extracts of cells containing expression plasmids for μPlg or miniPlg after induction with IPTG for 3 h. Lanes 1, MW std; lane 2, μPlg–IPTG induction; lane 3, μPlg + IPTG induction; lane 4, miniPlg–IPTG induction; lane 5, miniPlg + IPTG induction. (B) SDS–PAGE (with β-ME reduction) analysis of purified μPlg and miniPlg inclusion bodies. Lanes 1, MW std; lane 2, μPlg; lane 3, miniPlg.

fuged at 30,000 rpm, 4 °C for 30 min and characterized by A_{280} and non-reducing SDS–PAGE.

Refolding of denatured μPlg and miniPlg

The purified inclusion body preps were diluted in a 8 M urea buffer 8 M urea, 0.1 M Tris, 1 mM glycine, 1 mM EDTA, 10 mM β-ME, 10 mM dithiothreitol (DTT), 1 mM reduced glutathione (GSH), and 0.1 mM oxidized glutathione (GSSG), pH 10 with a final concentration of 2 mg/ml as an initial step just before rapid dilution refolding was initiated. Subsequently, for μPlg, this solution was rapidly diluted into 20 volumes 20 mM Tris, 0.2 M arginine, pH 9.5. For miniPlg, this solution was rapidly diluted into 20 volumes 20 mM Tris, 0.2 M arginine, pH 9.0. The final volume in the experiments ranged from 4 to 16 L. The solutions were allowed to refold over night at 18 °C (approximately 16 h) and then were transferred to a 4 °C cold room for an additional 48 h.

Purification procedure

For μPlg, purification was performed by first concentrating the refolding buffer to 0.5 L using tangential flow filtration through 3 Kvick Lab 10 K membranes (Amersham, Piscataway, NJ). The retentate was dialyzed against 20 mM MES, 10% sucrose, pH 6.5 overnight at 4 °C. The solution was centrifuged at 20 K rpm, 4 °C for 30 min, and the supernatant was loaded onto a 150 ml SP–Sepharose Fast Flow column (Amersham, Piscataway, NJ) equilibrated with three column volumes of buffer A: 20 mM MES, 10% sucrose, pH 6.5. Elution was performed using a linear gradient from 0% to 100% of buffer B: 20 mM Tris, 10% sucrose, and 1 M NaCl, pH 6.5 at 5 ml/min over 6 column volumes at room temperature with 10 ml fractions being collected. Peak fractions were combined and were then dialyzed against 20 mM citric acid, 10% sucrose, pH 3.1 overnight at 4 °C. The solution was concentrated to 10 ml using high pressure nitrogen filtration through a Millipore Amicon 10 K membrane (Millipore, Bedford, MA). The retentate was centrifuged at 30 K rpm, 4 °C for 30 min and loaded onto a Superdex 75 column (XK50 × 850-mm, Amersham, Piscataway, NJ) equilibrated with 2 column volumes of 20 mM citric acid, 7.5% sucrose, and 0.15 M NaCl, pH 3.1. The sample was eluted by running 1 column volume of 20 mM citric acid, 10% sucrose, and 0.15 M NaCl, pH 3.1 buffer at 4 °C with 10 ml fractions being collected. Peak fractions were identified by A_{280} and non-reducing SDS–PAGE and were then pooled and stored at −85 °C. The yield of isolated fractions was computed by molar extinction in 6 M guanidine, 38 mM NaPi, pH 6.5, where 1 mg/mL = $1.40 A_{280}$.

For miniPlg, purification was performed by first concentrating the refolding buffer to 0.5 L using tangential flow filtration using Kvick Lab 10 K membranes. The material was further concentrated to 20 ml using the Millipore Amicon N_2 pressure membrane concentrator, ultracentrifuged at 30 K rpm, 4 °C for 30 min and then loaded onto a Superdex 75 size exclusion column (XK50 × 850-mm, Amersham, Piscataway, NJ) equilibrated with 20 mM Tris, 0.2 M arginine, 0.15 M NaCl, pH 9.0 at 4 °C. Fractions (10 ml) were collected and analyzed by A_{280} and SDS–PAGE under non-reducing conditions. If necessary for "polishing", peak monomeric fractions (Mr ~ 38 kDa) were pooled and reconcentrated using the N_2 pressure membrane concentrator and loaded onto a Superdex 200 column (XK26 × 850-mm, Amersham, Piscataway, NJ) equilibrated with 20 mM citric acid, 10% sucrose, and 0.15 M NaCl, pH 3.1 at 4 °C. Peak fractions were identified at A_{280} and by non-reducing SDS–PAGE and were then pooled and stored at −85 °C. The yield of isolated fractions was computed by molar extinction in 6 M guanidine, 38 mM NaPi, pH 6.5, where 1 mg/mL = $1.54 A_{280}$.

Kinetic characterization of refolded molecules

Refolded μPlm and miniPlm were compared kinetically with commercially available plasmin (Sigma, St Louis, MO) at 21 °C using the commercially available plasmin chromogenic substrate S-2403 (L-pyroglutamyl-L-phenylalanyl-L-lysine-*p*-nitroaniline hydrochloride) (Chromogenix, Goteberg, Sweden). As an initial step, appropriate quantities of μPlg and miniPlg were diluted to 200 μg/mL in 100 mM Tris, 150 mM NaCl, and 0.01% Tween 20, pH 7.6. Urokinase was added at a molar stoichiometric ratio of 1:20 (urokinase:Plg) and incubated for 15 min at 37 °C to convert the zymogens to μPlm or miniPlm. Ten microlitre aliquots of μPlm, miniPlm, or plasmin were then rapidly mixed with 100 μL aliquots of S-2403 in 50 mM Tris, 150 mM NaCl, and 0.01% Tween 20, pH 7.6 at a range of concentrations of S-2403 (300–3000 μM) in quadruplicate using a titertek micropipetter in a 96-well microplate. The experiment was performed in the absence of fibrin or fibrinogen. The urokinase used to activate the plasmids does not catalyze amidolysis of S-2403. The A_{405} was measured at 15 s intervals for 5–15 min using a Spectramax 384 plate reader (Molecular Devices, Sunnyvale, CA). The raw data were converted to μM amounts and analyzed using a Hanes plot.

Amino acid sequencing

N-terminal amino acid sequencing was performed on individual protein bands excised from PVDF membrane protein blots (Millipore, Bedford, MA) at Molecular Biology Resource Facility, University of Oklahoma Health Sciences Center, Oklahoma City, Oklahoma.

Results

High level expression of μPlg and miniPlg as inclusion bodies

Fig. 1A depicts specific expression of μPlg and miniPlg expression in the presence and absence of IPTG. Fig. 1B depicts purified inclusion bodies for μPlg and miniPlg. The proteins in the inclusion bodies are highly purified before refolding is initiated. Although the data shown in Fig. 1A involved growth in LB broth and induction with IPTG, we routinely utilized the auto-induction system of Studier [9] for large scale preparation of inclusion bodies, since a 6-to10-fold higher yields of inclusion bodies were obtained. When the inclusion bodies were purified, as described in Materials and methods, a final yield of 500–600 mg of inclusion bodies was obtained per liter of ZYP-5052 expression medium for both μPlg and miniPlg.

Primary protein structure of μPlg and miniPlg expression constructs

Microplasmin produced from plasmin by alkaline treatment [3,4] results in a 31 amino acid A chain and a 230 amino acid B chain (Fig. 2A). The numbering system used in the figure uses the nomenclature for full length plasmin. Cysteines 548 and 558 each form essential disulfide bridges with cysteines in the B chain but amino acids 531–545 can be deleted without effect on function [5]. We constructed a recombinant form of single chain μPlg, designated as r-hu–μPlg in Fig. 2A, which was expressed as pET11a-μPlg as described in Materials and methods. The r-hu– μPlg is a fusion protein with an extra N-terminal 18 amino acid (Fig. 2A, underlined) for facilitating expression of inclusion bodies in *E. coli*. Treatment of r-hu–μplg with plasminogen activator such as urokinase results in cleavage not only at the authentic activation site, the peptide bond between arginine 561-valine 562 ($R^{561}\blacktriangledown V^{562}$), but also at an artificial site at the junction between the artificial amino terminus and the authentic amino acids of chain A. Cleavage at this site, indicated as a vertical pointing arrow in Fig. 2A, trims away all but a valine residue of the artificial amino terminus ($GR\blacktriangledown VA^{542}$) when the molecule is activated to μPlm. In a previous publication, Reich and coworkers [5] established that removal of the amino acids prior to serine 545 in Chain A was not detrimental to the catalytic properties of μPlm.

A schematic diagram of miniPlg is presented in Fig. 2B. The original miniPlg was generated by neutrophil elastase digestion of plasmin at the peptide bond between valine 441 and valine 442 [1,2]. Our recombinant miniPlg started from alanine 439, as indicated in Fig. 2B. N-terminal amino acid sequencing of the purified miniPlg indicates that the starting amino acid methionine is removed during expression (data not shown).

Refolding and purification of μPlg and miniPlg

Microplasminogen has six disulfide bridges and miniPlg has nine disulfide bridges. Therefore, it is not surprising that there would be a significant amount of incorrectly formed disulfides and multimeric forms of the protein generated during the refolding process. Before arriving at final refolding conditions, described in Materials and methods section and in Table 1, we performed a small scale "refolding screening" process consisting of a matrix in which more than 40 refolding conditions were examined (data not shown). Criterion for selection in the initial screening included lack of visible protein precipitate during refolding or in subsequent concentration steps for SEC, and elution of protein peaks from the SEC with a mobility identical to refolded monomeric forms of the protein. SEC is used as an important tool in post-refolding purification because multimeric, unfolded, soluble forms of the protein can be easily separated from refolded monomer. Ultimately, the refolding conditions summarized in Table 1 were selected because they generated the highest percent of monomeric form of μPlg or miniPlg in the SEC chromatographic step. They also presented a favorable activity profile when the proteins were activated to either μPlm or miniPlm and examined in a functional assay (discussed below). Representative chromatograms of isolated μPlg and miniPlg are presented in Fig. 3A and C, respectively. Overall,

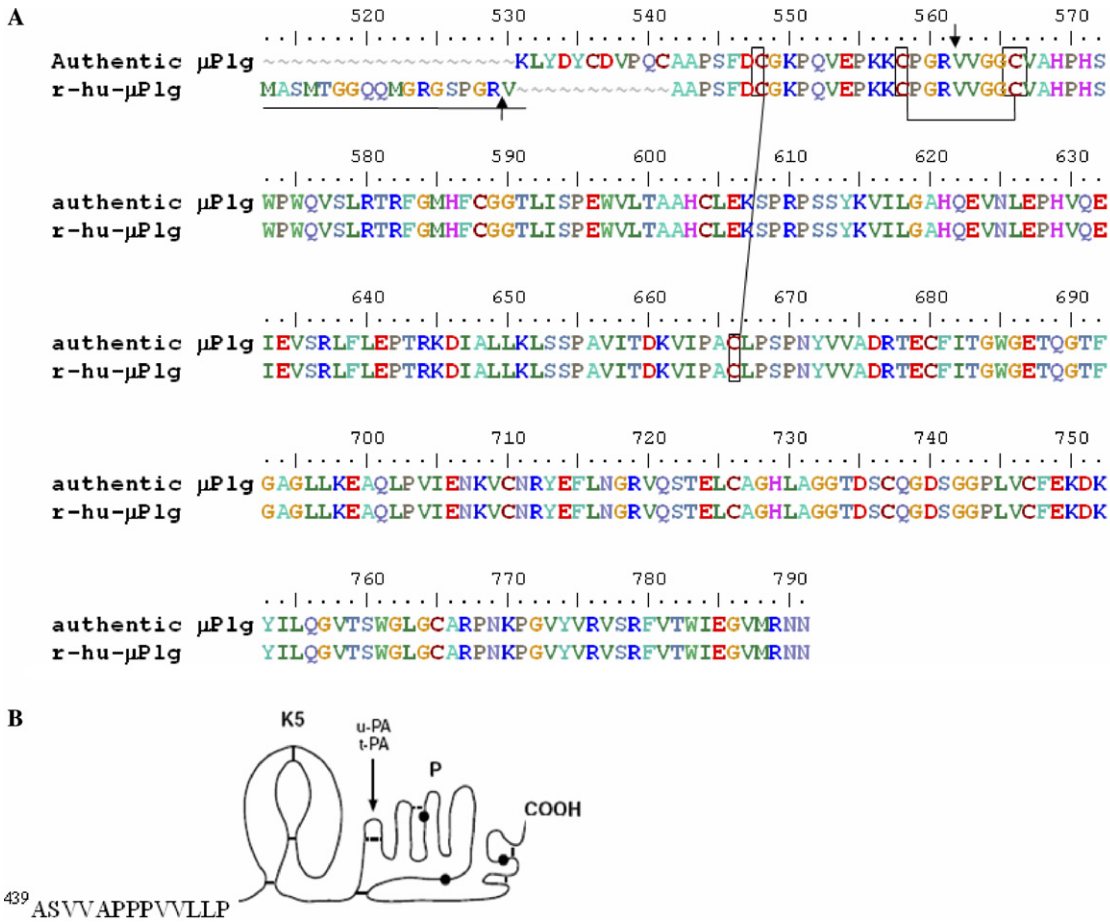


Fig. 2. (A) Amino acid sequence of authentic microplasmin as originally generated after pH 11 treatment [3,4] and the recombinant construct (r-hu–µPlg) used in this paper. The numbering system is derived from full length plasmin. Arrow pointing downward indicates the authentic urokinase activation cleavage site R^{561}▼V^{562}, while the upward pointing arrow indicates an artificial urokinase cleavage site engineered to trim away the molecule of inauthentic amino acids in plasmid construct following activation. The inauthentic amino acids are underlined by a single line. Only an inauthentic valine will remain. Boxed cysteines indicate essential disulfide bridges between chain A and chain B after activation. (B) Schematic diagram of recombinant miniplasminogen construct used in this study.

approximately 10% of total protein refolded from inclusion bodies, estimated to be >90% pure at the outset, was isolated in the final monomeric refolded form at >98% purity (Table 1, Fig. 3B and D). Because of the very high yield of inclusion bodies using the T7 promoter and the auto-induction expression system of Studier [9], we estimate that we can isolate at least 50–60 mg of purified, fully function, monomeric µPlm or miniPlm from each liter of expression culture using Fernbach shaker flasks.

Auto-activation

After isolating monomeric correctly refolded µPlg and miniPlg, it was discovered that both of these molecules have a tendency to auto-activate to µPlm and miniPlm, respectively, if the isolated fractions were left at neutral or alkaline pH at 4 °C over several days. Unlike the zymogen form, the auto-activated material displayed full enzymatic activity. The activation occurs without the addition of

Table 1
Summary of refolding protocol for µplasminogen and miniplasminogen

Protein	µPlasminogen	Miniplasminogen
Refolding buffer	20 mM Tris, 0.2 M arginine, 0.4 M urea, 0.05 mM GSH, 0.005 mM GSSG, 0.5 mM β-ME, and 0.05 mM EDTA pH 9.5	20 mM Tris, 0.2 M arginine, 0.02 mM PMSF, 0.4 M urea, 0.05 mM GSH, 0.005 mM GSSG, 0.5 mM β-ME, and 0.05 mM EDTA, pH 9.5
Refolding concentration	0.1 mg/mL	0.1 mg/mL
Refolding time	Overnight at 18 °C, then 48 h at 4 °C	Overnight at 18 °C, then 48 h at 4 °C
Total protein monomer yield after 1st step of purification	20–25% after SP IEX chromatography	15–20% after Superdex 75 SEC
Total protein monomer yield after 2nd step of purification	10% after Superdex 75 SEC	10% after Superdex 200 SEC

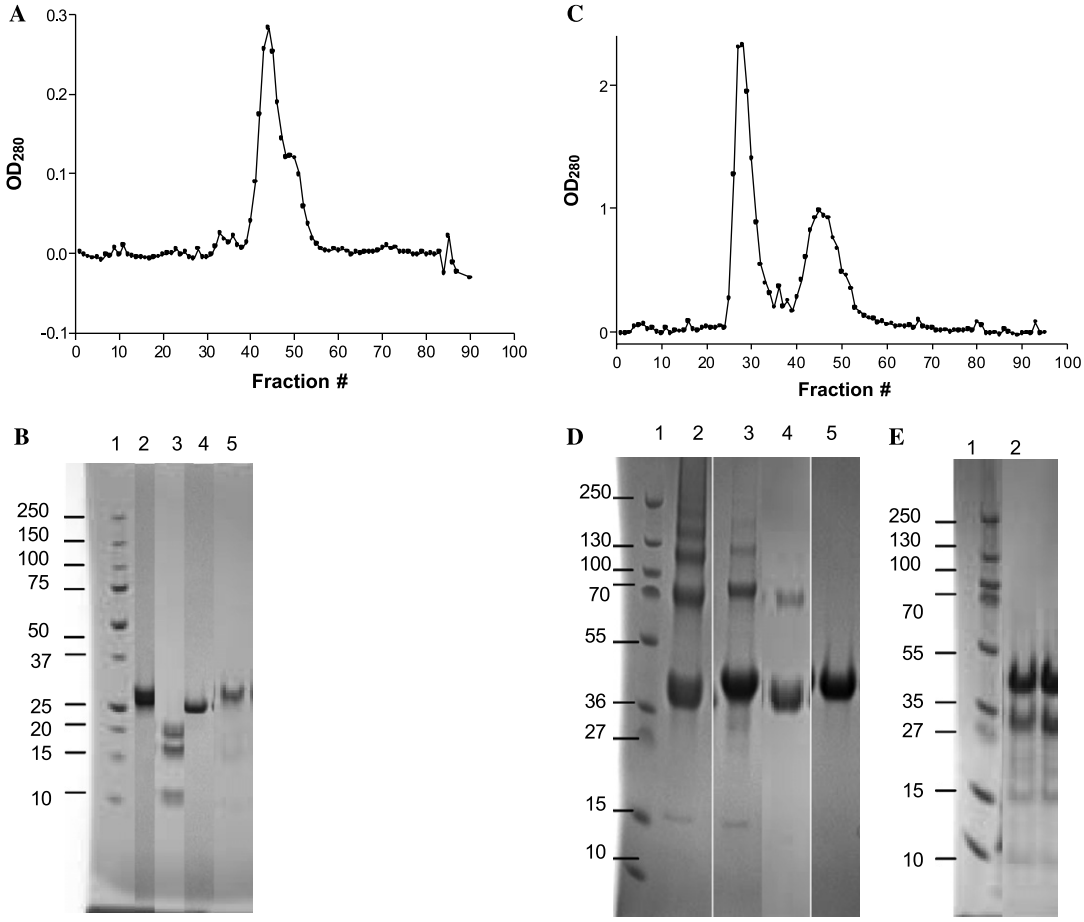
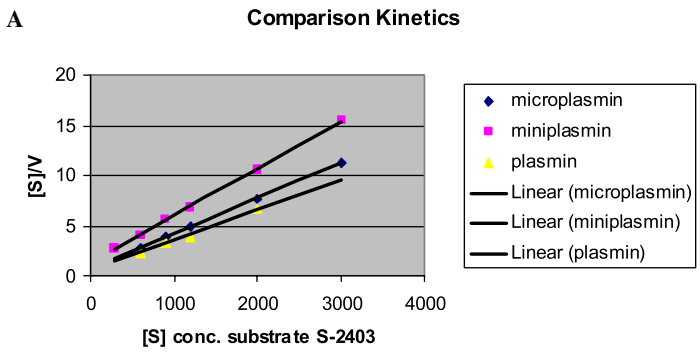


Fig. 3. (A) Superdex 75 chromatogram of refolded μPlg. (B) SDS–PAGE of μPlg fractions isolated in Fig 3A. Lanes 1, MW std; lane 2, F 42 non reduced; lane 3, F42 reduced; lane 4, F50 non-reduced; lane 5, F50 reduced. Comment: non-reduced, non-activated (full length) μPlg has an apparent mobility faster (Mr ~ 29 kDa) than the reduced form (Mr ~ 32 kDa) (lane 4 vs. lane 5). The lower molecular weight bands in the reduced lane 3 represent auto-activated, fragments as discussed in text. (C) Superdex 75 of refolded miniPlg. The elution peak on the left consists of multimeric unfolded forms of miniPlg while the protein peak on the right consists primarily of miniPlg monomers. (D) SDS–PAGE of Superdex 75 chromatography in Fig. 3C Lanes 1, MW std; lane 2, non-reduced F28, lane 3, reduced F28; lane 4, non-reduced F45; lane 5, reduced F45. (E) A sample of purified miniPlg stored in alkaline pH buffer(20 mM Tris, pH 9.0, 0.2 M arginine, and 0.15 M NaCl) auto-activated after storage at 4 °C for 1 week. Lanes 1, MW std; lane 2, reduced sample of miniPlg.

plasminogen activator, urokinase. The actual auto-activation could be detected clearly after SEC by SDS–PAGE if samples were reduced with β-ME. In the case of μPlg, this was clearly demonstrated when material isolated after a first column purification step (IEX) was left at neutral pH for several days before running a second purification step (SEC column). Analysis of the SEC column (Fig. 3A) fractions by SDS–PAGE (Fig. 3B) indicated separation into 2 peaks—one of auto-activated material (Fig. 3B, lanes 2 and 3) and one of unactivated zymogen (Fig. 3B, lanes 4 and 5). The auto-activated material was not visible without reduction because of the interconnection by disulfide bonds (Fig. 3B, lanes 2 and 4). Interestingly, unactivated reduced μPlg displayed a reduced mobility on the gel relative to its non-reduced state (Fig. 3B lanes 5 vs. lane 4), probably because the polypeptide backbone of the protein linearized after reduction. N-terminal amino acid sequencing of PVDF protein blots from the SDS–PAGE bands confirmed that the auto-activation occurred at $R^{561}\blacktriangledown V^{562}$, which is the authentic activation site for plasminogen activator. We ultimately discovered that auto-activation could be prevented completely, if the pH was lowered immediately either by dialysis or by adding solid sodium citrate and sucrose (final buffer concentration: 20 mM citrate, pH 3.1, 10% sucrose). This observation parallels that made by Jensen [10] three decades ago for stabilizing full length plasmin. By including the rapid addition of sucrose and citrate into the isolated fractions in our protocol as described in Material and methods, we no longer observe the auto-activation of μPlg presented in Fig. 3B.

Using similar method for μPlg purification, we have also incorportated the citrate/sucrose buffer into our separation procedure when purifying the refolded miniPlg. As is demonstrated in Fig. 3C, during the first step of chromatography, SEC, two peaks were isolated. The first peak contained soluble, misfolded, high molecular weight multimers of miniPlg (Fig. 3D, lanes 2 and 3) while the second peak consisted primarily of the monomeric form (Fig. 3C, lanes 4 and 5). Solid citrate and sucrose were added to the column fractions immediately after isolation to prevent auto-activation.



B

Molecule	Km (μM)	Kcat (min^{-1})	Kcat/Km ($min^{-1}\mu M^{-1}$)
μplasmin	192	2621	13.65
miniplasmin	265	2814	10.62
plasmin	165	4409	26.72

Fig. 4. (A) Hanes kinetic comparison of refolded μPlm, miniPlm, and commercially purchased plasmin using amidolytic chromogenic substrate S-2403. (B) Summary of the kinetic parameters obtained in Fig. 4A.

The purified material could be stored at 4 °C in citrate buffer at pH 3.1 for more than 30 days without evidence of auto-activation, and could also be frozen without loss of activity. However, if the material was re-elevated to neutral pH, auto-activation would occur (Fig. 3E, lane 2).

Kinetic characterization of μPlm and miniPlm

Refolded μPlm and miniPlm were compared kinetically with commercially available plasmin at 21 °C using plasmin chromogenic substrate S-2403. As described in detail in Materials and methods, a fixed concentration of μPlm, miniPlm, or plasmin was mixed with a range of concentrations of S-2403 (300–3000 μM). As summarized in Fig. 4, the kinetic parameters for the three molecules were relatively similar. Plasmin had the lowest K_m (165 μM) and the highest K_{cat} (4409 min^{-1}); its catalytic efficiency was about 2× higher than μPlm and about 2.6× higher than miniPlm toward the synthetic substrate. These initial data establish that the active site conformation of the serine protease domain of the three molecules is very similar, validated the refolding methodology. However, it is well known that fibrin interacts with the kringle domains of plasmin and miniPlm, significantly increasing the K_{cat} [11–13] relative to μPlm. We have not yet compared these 3 enzymes using fibrin as a competing substrate, as a full scale kinetic characterization of μPlm, miniPlm, and plasmin in the presence of fibrin and/or fibrinogen is beyond the scope of this report but will be discussed in the Discussion section.

Discussion

Recently published *in vivo* animal studies indicate that plasmin and its des-kringle derivatives μPlm, and by implication, miniPlm, may be attractive therapeutic options for the treatment of PAOD and ischemic stroke. Due to their short serum half-lives—on the seconds time scale [8]—and longer half lives when bound to fibrin surface—two to three orders of magnitude longer than in serum circulation [14]—they represent an attractive alternative to treatment with plasminogen activators when used in conjunction with catheters that permit localized delivery directly to a clot. Their action is restricted exclusively to the site of delivery, whereas tPA, with a half life upwards to 15 min, can promote systemic bleeding, distal clot dissolution, and the risk of hemorrhaging. Plasmin, however, has fragile, complex structures and has never been produced recombinantly at a pharmaceutical scale—their lability has hampered commercial development. Nagai and coworkers [8] have demonstrated the feasibility of producing large scale amounts of μPlg using a *P. pastoris* expression system. μPlg has also been produced utilizing a baculovirus expression system [5], but with relatively low yield.

In this report, we demonstrate that large quantities of μPlg and miniPlg can be produced as inclusion bodies in *E. coli*, and refolded to functionality. We have used the auto-induction protein expression system developed by Studier [9] in conjunction with expression plasmids containing an T7 promoter controlled by a *lac operator*, to generate high-level expression of inclusion bodies in a shaker flask expression system. The quantity of μPlg or miniPlg produced as inclusion bodies exceeds that using a LB broth/IPTG inducible T7 expression system by 6- to 10-fold in these specific cases. We produced 500–600 mg/L of purified inclusion bodies routinely for μPlg and miniPlg, and the final recovery of functionally active monomeric protein after refolding and purification—about 10%—represent at least 50 mg of recombinant products.

To our knowledge, this report is the first successful production of functional miniPlg in any recombinant expression system. Previously, investigators had successfully produced kringle 5 alone isolated from *E. coli* inclusion bodies [15] or as a soluble protein isolated from *P. pastoris* [16]. μPlg had been produced successfully in an insect cell baculovirus expression system [5] and in *P. pastoris* [8]. To our knowledge, this is the first wild-type miniPlg refolded from *E. coli* system in sufficient yield to initiate animal studies on its efficacy, pharmacokinetic properties, and toxicity.

An important observation made during the refolding and purification of μPlg and miniPlg were their tendency to auto-activate from zymogen to active form by cleavage at the correct site for physiological activation—$R^{561}\blacktriangledown V^{562}$ (Fig. 2). This phenomenon occurred in the *absence* of plasminogen activators urokinase or tPA. The plasminogen activators are exquisitely sensitive in their site specificity, but plasmin, although

more specific than trypsin, is relatively non-specific at cleaving proteins, and therefore it is not surprising to observe this occurrence. Under conditions of neutral or alkaline pH, auto-activation was observed to occur over several days time scale at 4 °C. To cope with this inherent instability, we have utilized a method developed by Jensen [10], initially developed to stabilize plasmin solutions, in which low pH citrate buffers (e.g., pH 3.1) and sucrose were reported to be effective stabilizers, and applied it to our μPlg and miniPlg purification and stabilization protocols.

We have completed a partial characterization of the functional properties of μPlm and miniPlm, comparing the serine protease activity to commercially available plasmin. As was reported in Fig. 4, when using the amidolytic chromophore substrate S-2403 (L-pyroglutamyl-L-phenylalanyl-L-lysine-*p*NA), the K_m values for plasmin, μPlm, and miniPlm were in a range of 165–265 μM (Fig. 4). The catalytic efficiency, K_{cat}/K_m, for plasmin was 2.6× higher than μPlm and 2× higher than miniPlm. Although this validates our refolding protocol for the serine protease domain of μPlm and miniPlm, further studies are necessary to validate the functional refolding of the kringle 5 domain of miniPlm, which possesses a lysine binding domain which is important in interactions with fibrin and α_2-antiplasmin (AP). Interaction of fibrin with the kringle domains of plasmin significantly increases the catalytic efficiency of the enzyme [11]. Although, we have established that stoichiometric quantities of AP inhibits both μPlm and miniPlm (data not shown), millisecond stopped flow kinetics are required to make a careful comparison [17]. A simple test of binding of miniPlg or miniPlm to a lysine affinity column is not possible, because, unlike kringles 1–4 of plasmin, kringle 5 does not bind to this affinity column [18]. Fairly complicated experiments using fibrin in competition with S-2403 (both substrates) will be required to elucidate the functional role of kringle 5 on substrate–enzyme kinetics for miniPlm in comparison to μPlm or plasmin. Using μPlm and miniPlm produced from serum-derived plasmin, Machovich [11] and coworkers reported that in the presence of fibrin, μPlm and plasmin had nearly identical $K_{m's}$, but the overall catalytic efficiency K_{cat}/K_m of plasmin was 7.3× higher; when compared to miniPlm, the K_{cat}/K_m of plasmin was 2.0× higher. This implies that, *in vivo*, higher doses of miniPlm or μPlm will be required to achieve equivalent rates of fibrin degradation by plasmin. If higher doses are actually required, particularly for μPlm, bleeding complications may arise. Such considerations may be important in selecting miniPlm over μPlm for therapeutic development. However, these studies are beyond the scope of this report.

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